UMHACKATHON 2026 · DOMAIN 1 · AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

Product Requirements.

The product the platform is, the users it serves, the scope it commits to for the UMHackathon 2026 submission.

ONBOARD

POWERED BY Z.AI GLM · REGULATED ONBOARDING ENGINE

PRD · PRODUCT REQUIREMENTS APRIL 2026 SUBMISSION RELEASE

1. Problem Definition & Purpose

Opening a new building is slow, paper-heavy, and local. Ownership certificates, fire-safety approvals, zoning clearances, floor plans, safety reports — each submitted to a different office, each with its own format, each city wanting it slightly differently. Developers file paperwork they do not fully understand and wait weeks for feedback; cities get a flood of non-standard submissions; the public has no good way to discover what is new, approved, or available.

The bottleneck is not bureaucracy as a concept. It is the **absence of a system that can read what's submitted, check it against what a city requires, and keep the process moving without waiting on humans to re-read every form**.

Purpose. Opus Magnum turns regulated building onboarding into a single guided workflow, powered end-to-end by an AI that reads documents, verifies them against city-specific criteria, reconciles contradictions, and publishes the building as a public, explorable listing.

Larger ambition. Buildings are the concrete demo. The engine underneath is a **general-purpose adaptive, multi-modal, AI-verified onboarding platform** — the same template model applies to supplier qualification, license renewal, KYC, regulated-professional registration. We ship buildings because the pain is concrete and visual; the pattern is reusable.

2. Target Users & User Stories

Three roles, one engine, one knowledge base.

2.1 Admin — the city's configuration point

Who. City-office workflow owner (municipal digital-services manager, or equivalent). **Core job.** Define the onboarding process a city requires: which forms, which documents, which verification criteria, in what order.

Stories.

- *As an admin, I can describe my city's requirements in one sentence and get a GLM-drafted starter template I can edit, so I don't start from a blank page.*
- *As an admin, I can ask a tutor assistant to explain why each step exists and what a field should capture, so I can author confidently without a manual.*
- *As an admin, I can publish a template and have every new onboarding in my city run through that exact sequence, so that submissions become comparable across buildings.*

2.2 Building owner — the workflow user

Who. Developer, property-management lead, or their appointed submitter. **Core job.** Onboard a specific building by running that city's template to completion.

Stories.

- *As an owner, I can see the whole onboarding process as a live graph, so I always know what is done, what is running, and what is waiting on me.*
- *As an owner, when I upload a compliance document, I can watch the AI process it in real time — received, parsed, extracted, verified — and see a plain-language pass or fail verdict with evidence.*
- *As an owner, when the AI flags a contradiction between documents, I can see exactly which values disagree and correct them without a round-trip to a reviewer.*
- *As an owner, when all steps pass, my building is published automatically — the paperwork and the public listing come from the same submission.*

2.3 Visitor — the public audience

Who. Anyone curious about a published building — prospective tenants, journalists, city residents, neighbouring businesses. **Core job.** Discover a building and ask grounded questions about it.

Stories.

- *As a visitor, I can browse a directory of published buildings and open one to see a 3D view and a chat panel.*
- *As a visitor, I can ask the assistant anything about the building, and it answers only from what the owner submitted — with citations to the source document or profile field, never invention.*

3. Originality, Innovation & Value Realisation

3.1 What is original

The workflow is data, not code. A new city is a new JSON template stored in Postgres. The same engine runs every template. There is no per-city codebase. No YAML rule files. No "compliance-rules microservice." The admin's template *is* the rule set.

Compliance docs vs content docs are explicit first-class types. Compliance docs feed pass/fail verdicts and are never shown publicly. Content docs feed the building's profile and ground the visitor assistant. The same upload workflow handles both, the system never conflates them, and the owner submits everything once.

Every GLM call is logged for audit. Each step's `decision_log` captures the prompt, the tool called, the model's arguments, and the duration. This is the drift net for AI behaviour and the substrate for future human-in-the-loop review of AI judgements.

3.2 Why it matters

- **For admins:** a city that wants to onboard buildings for the first time can be live in minutes, not quarters. The marginal cost of a new jurisdiction is one template.
- **For owners:** the feedback loop collapses from weeks to seconds. A rejection cites the exact reason; corrections go straight back into the graph.
- **For cities:** submissions become uniform and auditable. Every decision trail is preserved.
- **For the public:** discovery becomes first-class. A building's page is not a document dump — it is a conversable, grounded listing.

3.3 The "AI at the centre" invariant

If GLM is removed, five things stop working:

1. **Template drafting.** Admins lose the GLM-generated starter template and the tutor assistant; only manual authoring remains.
2. **Document parsing & fact extraction.** Raw PDFs / images never become structured fields.
3. **Compliance verification.** Plain-English criteria cannot be applied to a document without reading it.
4. **Cross-document reconciliation.** Contradiction surfacing requires reasoning over multiple extracted outputs.
5. **Visitor assistant.** Every grounded answer depends on reading the building's content documents.

Form rendering survives. Nothing else does. This is the architectural litmus test judged in §6.

4. Feature Prioritisation & MVP Scope

4.1 In scope for MVP

AREA	FEATURE	STATUS
Admin	City CRUD	Shipped
Admin	Template editor with primitive-specific config	Shipped
Admin	`draft_with_ai` (GLM drafts full template from one-line brief)	Shipped
Admin	Template assistant chat (SSE-streamed, explains fields)	Shipped
Owner	Building create + city selection	Shipped
Owner	Workflow canvas (React Flow) with live step status	Shipped
Owner	Dynamic form renderer for `collect_form`	Shipped
Owner	Drag-drop upload with live per-file processing cards	Shipped
Owner	Compliance verification loop (re-upload on fail)	Shipped
Owner	Content extraction into the building profile	Shipped
Owner	Cross-document check with severity grading	Shipped
Owner	Human-review summary + publish	Shipped
Owner	Per-step decision-log drawer (every GLM call visible)	Shipped
Visitor	Published directory	Shipped
Visitor	3D scene (real GLB apartment models + scene_config overlay)	Shipped
Visitor	Grounded assistant chat (SSE-streamed, profile + content docs)	Shipped
Platform	Six workflow primitives, composable per city	Shipped
Platform	LangGraph StateGraph per run with Postgres/SQLite checkpointer	Shipped
Platform	OpenAI-compatible integration with ILMU `ilmu-glm-5.1`	Shipped
Platform	`decision_log` persisted per step	Shipped

4.2 Out of scope — flagged explicitly

- **Real auth.** Mocked session: one seeded admin and one seeded owner. Drop-in JWT is the designed upgrade path; this was cut to focus engineering on the GLM-critical loop.
- **Payments / transactions.** Not in the onboarding value chain for MVP.
- **Multi-tenant isolation & row-level security.** Single-tenant demo.
- **Production observability & alerting.** Dev-grade logging only.
- **Second jurisdiction.** The engine is config-driven; only Shah Alam is authored for the submission. Adding Kuala Lumpur would be one JSON file.
- **Vision documents.** `ilmu-glm-5.1` does not accept image inputs (verified empirically — see §7 Risks). Image documents go through OCR (`pytesseract`) before GLM sees them.

4.3 Prioritisation rationale

The demo judged here is three minutes long. Every shipped feature earns a beat in that demo. Features that do not appear in the demo script (real auth, payments, observability) were cut so the GLM-critical path could be built deeply rather than broadly.

5. Success Metrics (MVP)

METRIC	TARGET	HOW MEASURED
Live GLM calls per run	≥ 12 per full Shah Alam template	`decision_log` count per `workflow_run`
Compliance verification accuracy	≥ 9 / 10 on labelled fixtures	`tests/test_upload_compliance.py` + GLM eval set
Verification rejection with evidence	100% of failing cases cite offending text	Decision-log inspection
End-to-end demo time	≤ 3 minutes live	Timed rehearsal
Template authoring time	≤ 2 minutes from blank to publish via `draft_with_ai`	Manual measurement

6. GLM-Centrality Validation

Every one of the five critical surfaces calls GLM at runtime, each logged under `step_runs.decision_log` with its tool name, arguments, and timing. A reviewer who stubs `GLMClient` to raise immediately would see the workflow freeze at step 1, template drafting fail, and the visitor assistant return an error. The litmus test passes.

7. Risks & Mitigations

RISK	IMPACT	MITIGATION
`ilmu-glm-5.1` rate limits trigger during demo	Mid-demo stall	Exponential-backoff retry in `GLMClient._chat`; demo path uses ≤ 15 calls end-to-end
Vision not supported on GLM 5.1	Image docs can't be read directly	OCR-only fallback (`pytesseract`), flagged in decision log; baseline verified
GLM returns malformed tool arguments	Step stalls	`call_tool` catches `json.JSONDecodeError`, returns `{_error}` so the UI can show a retry affordance
Owner submits a scanned PDF with no text layer	Silent empty extract	`pdf.py` falls back to `pdf2image` + `pytesseract`; decision log marks `ocr_used=true`
Template becomes invalid mid-edit (unknown primitive)	Run crashes	`compile_template` raises `ValueError("unknown primitive: …")` before the graph executes
Checkpointer DB corruption	Run cannot resume	SQLite file scoped to `var/langgraph.sqlite`; deletable without data loss (state also mirrored in `workflow_runs.state`)

8. Future Work (not in this submission)

- Second city authored (Kuala Lumpur) to prove config-only expansion.
- Real multi-tenant auth + per-seat isolation.
- Embeddings-based RAG for very large content corpora (fallback if single-building content exceeds 200k tokens).
- Vision-capable model upgrade when ILMU adds one to our plan.
- City-side reviewer surface that spot-checks GLM verdicts.

Source: `implementation.md` §1–19 · `idea-presentation.md` · this document is the canonical product statement for the submission.